VetAMR-AI Technical Report

Team Kizuna

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Executive summary

VetAMR-AI implements a reproducible workflow for extracting veterinary antibiotic product details and checking package sales against reference concentrations. It combines a curated catalog, text extraction and harmonisation, active-substance calculations, descriptive analysis, and a browser interface.

The catalog contains 329 records across 62 primary ingredient groups. The 26 most frequent groups cover at least 80% of its records. A development evaluation matches all six annotated fields in each of two illustrative English leaflets. Automated verification passes 23 Python tests and four static-adapter checks. These results establish reproducible behaviour on the included data; broader scientific validity requires independent source verification and a larger evaluation corpus.

Problem and intended use

Challenge 8 calls for structured veterinary antibiotic reference data that can support verification of national antimicrobial use submissions. Product descriptions often express packaging, strengths, administration routes, and species in different forms. VetAMR-AI organises those fields and exposes the intermediate records so that a reviewer can inspect a proposed match and its resulting mass calculation.

The intended workflow is supervised: extract product information, review ambiguous fields, compare a reported product with the catalog, and calculate ingredient mass when the units are supported. The application flags unmatched products and unsupported dimensions instead of assigning a nominal concentration.

Data and methods

The catalog is reconstructed from the included seed and curation tables in vet_antibiotic_atlas/data/. Many records carry authority names and registration identifiers. Their source URLs, document archives, retrieval dates, reuse permissions, and independent review status are incomplete. They constitute a curated research catalog rather than a verified census of current authorisations.

The default extractor uses ingredient aliases, regular expressions, context rules, and unit, route, and species normalisation. PDF processing extracts embedded text; validated OCR is not included. An optional LLM adapter is available but is outside the reported evaluation. Completeness scores are heuristics and are not calibrated probabilities of correctness.

Repository analysis assigns each record to its primary ingredient group, ranks groups by record count, and uses country spread to break ties. The priority subset is the smallest ranked prefix reaching 80% of records. Secondary ingredients in combination products do not contribute separately to this ranking.

Evaluation design

scripts/evaluate_submission.py reads the two included illustrative English leaflets and evaluates actual parser predictions against committed annotations. The six fields are product name, ingredient set, ingredient-concentration pairs, package-volume set, route set, and target-species set.

Names are compared after case and whitespace normalisation. Set comparisons are exact, with numerical values rounded to six decimal places. Field accuracy is the number of exact document-field matches divided by all evaluated pairs. Precision, recall, and F1 are also calculated for the ingredient, package, route, and species sets. Every expected value and prediction is retained in evaluation/results.json, together with document hashes and regenerated catalog statistics.

Both leaflets were used during debugging. They are development fixtures, not a held-out or independently annotated corpus. Their provenance has not been verified against archived official source documents.

Results

Measure	Reproduced result	Interpretation
Catalog records	329	Included curated records
Primary ingredient groups	62	Groups used for frequency analysis
Groups reaching 80% record coverage	26 of 62, or 41.9%	A shortlist for review within this catalog
Exact extraction field matches	12 of 12 across two leaflets	Development consistency only
Python checks	23 passed	Unit and API behaviours exercised by the suite
Static-adapter checks	4 passed	Filtering, sample replay, input rejection, and CSV generation

The catalog result is not an empirical demonstration of an 80/20 law: 41.9% of groups are needed to reach the 80% threshold. It also does not establish sales volume, national consumption coverage, global prevalence, or audit confidence.

AMU calculation and uncertainty handling

For liquid formulations, active mass in kilograms equals the number of packs sold multiplied by pack volume in millilitres and concentration in milligrams per millilitre, divided by 1,000,000. For solids, the corresponding inputs are pack mass in grams and concentration in milligrams per gram. Supported combination ingredients are calculated separately.

For example, 1,000 packs of 100 mL containing 140 mg/mL amoxicillin and 35 mg/mL clavulanic acid produce 14.0 kg and 3.5 kg respectively. Unmatched products and unsupported concentration or package dimensions are flagged for review. Their uncalculated mass is excluded from totals; a displayed zero for

such a row is not evidence of zero antimicrobial use. A batch total may therefore be incomplete.

Salt-to-base factors, IU conversions, fuzzy product matches, and uncertain source fields need further substance-specific validation. A catalog match alone does not confirm current approval in the reporting jurisdiction.

Additional functions and deployment

The biomass module provides exploratory head-count and assumed-weight calculations, with dose-based indicators from an included reference table. It does not implement all population categories, production flows, or trade adjustments needed for a validated national biomass methodology. Exposure tiers and dose fallbacks require authoritative review. ANIMUSE-style exports are prototype mappings, not certified reporting schemas. SHA-256 checksums support change detection relative to a stored hash; they do not authenticate a source or signer.

Registry connectors return offline fixtures by default. The experimental FDA request can be enabled explicitly; the EMA feed remains fixture-based. Screening rules identify records for expert review and do not determine legal authorisation.

The Python application processes supported user inputs. The public GitHub Pages site offers a catalog, charts, and exact sample replays. Unsupported actions are rejected with instructions to use the Python application.

Reproduction

Follow the README to install dependencies, run the tests, regenerate the evaluation, export the static snapshot, and validate the website. Evaluation and export use isolated databases seeded from the included tables. The GitHub verification workflow provides an additional reproducibility check. Passing tests do not imply complete test coverage or establish scientific generalisation.

Priorities for further validation

The next study should assemble a licensed multilingual corpus with independent annotations, preserve field-level source evidence, and measure extraction errors by language, source, and field. Veterinary specialists should validate strengths, salts, combinations, routes, and species. Reporting examples should be reconciled with current WOAH schemas before a supervised national pilot measures correction time and agreement with independently checked ingredient quantities.

References and supporting evidence

WOAH Challenge 8

WOAH assessment criteria

WOAH submission instructions

Private submission repository

VetAMR-AI application

Supporting evidence is retained in evaluation/annotations.json and evaluation/results.json. Catalog construction is implemented in vet_antibiotic_atlas/data/; extraction, standardisation, analytics, and verification are organised into the corresponding package directories.